

Vibration Instruments and Setup

MENG 4050

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Machine Health Monitoring Vibration Instruments and Setup

01. Sensors for VCM

02. Signal Acquisition Fundamental

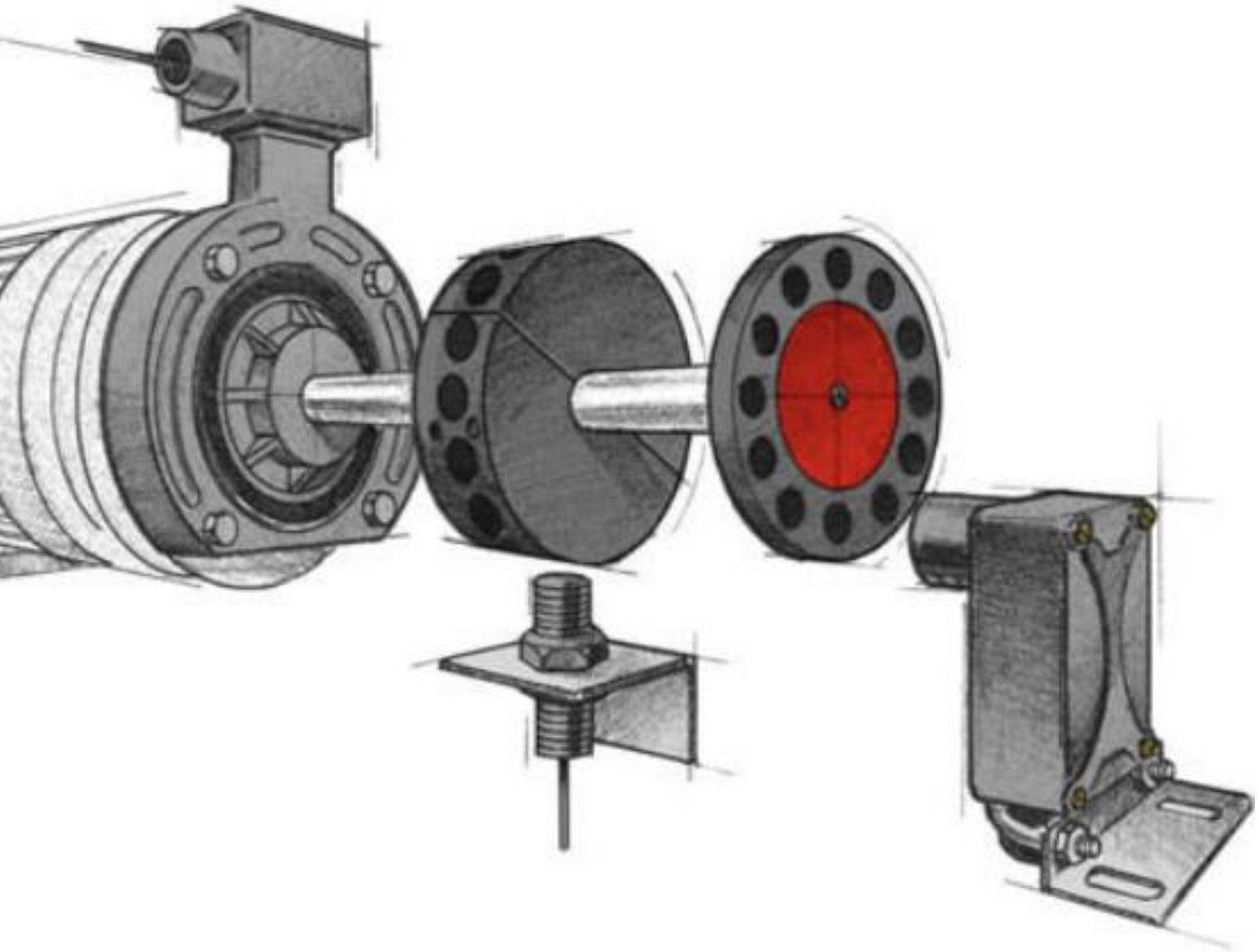




Section One

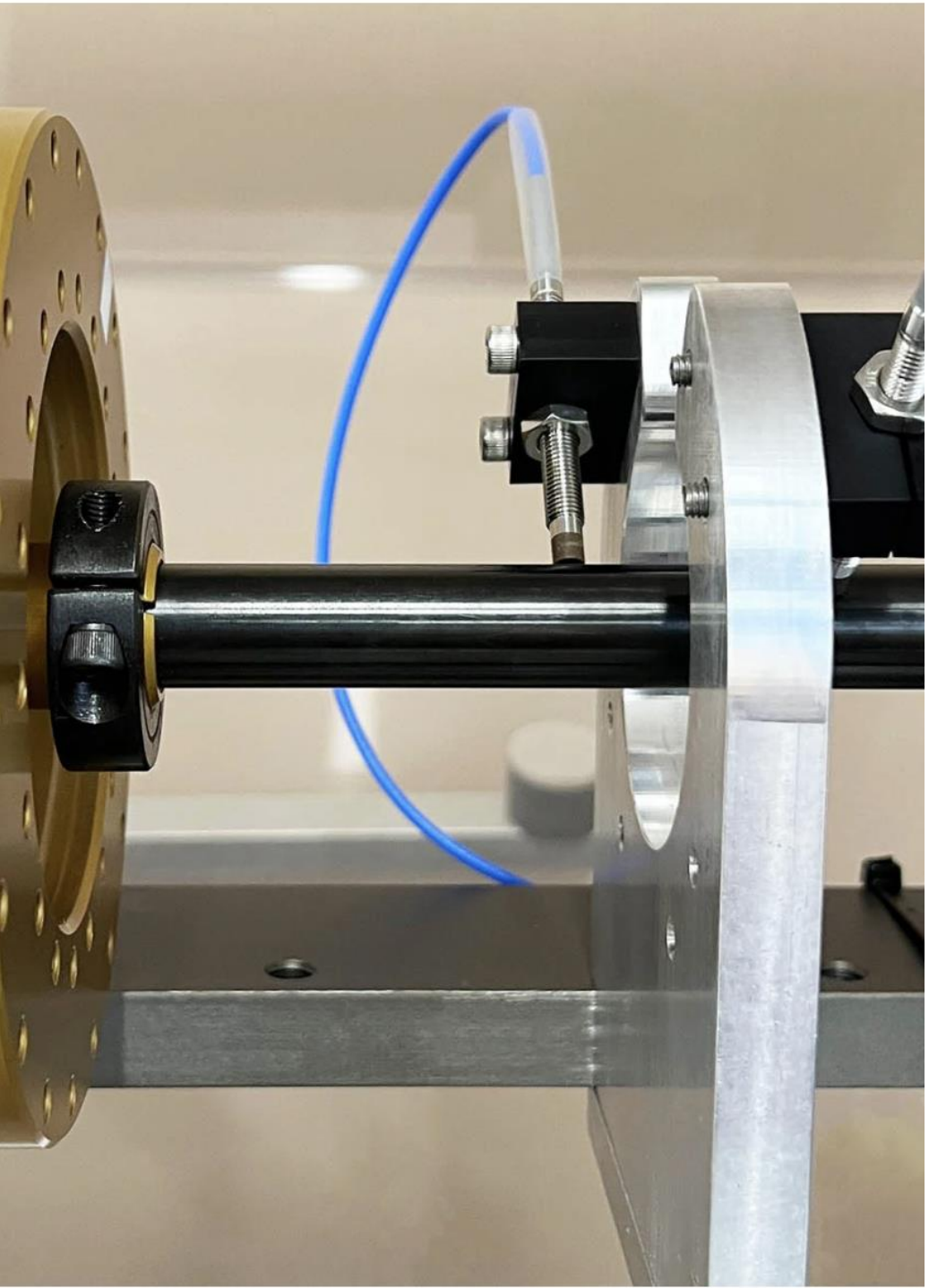
We will cover Sensors for VCM

- Sensors
- Sensor Mounting



SENSORS AND THEIR MOUNTING APPROACH

- Displacement Sensor
- Velocity Sensor
- Acceleration Sensor
- Tacho Sensor



Displacement Sensor

Proximity probe (or Eddy current probe) is used for measuring the relative displacement of a moving metal object relative to the sensor position.

- A proximity probe assembly consists of a probe containing a coil.
- The coil is excited by an input current and voltage at a high frequency, typically 1 MHz or above.
- The high frequency of excitation is responsible for inducing eddy currents on the surface of the target object.
- The eddy current alters the induction of the probe coil, and this change translate to the voltage output, which is proportional to the distance between the probe and the vibrating object.

This sensor is used for measuring shaft relative displacement with respect to the bearing pedestal in rotating machinery.



Velocity Sensor

Seismometer is a sensor to measure the vibration velocity and commonly used to measure the seismic events.

- Laser velocity sensor is used for measuring vibration velocity.
- works on the principle of the Doppler effect.
- The laser beam (pointer) from the sensor has to be focused on the vibrating object.

Advantages of the laser vibrometer (sensor) are

- measurement is possible from distance (no accessibility constraints).
- Non-contact nature ensures that the weight of the sensor is not added to that of the vibrating object.
- a single sensor can replace several contact type vibration sensors, by simply scanning the vibration of an entire area of the object.
- vibration measurement on small vibrating objects (up to micro-level) is also possible.

Acceleration Sensor

The piezoelectric accelerometers are the most popular and most commonly used.

It consists of a mass (called seismic mass) mounted on a piezoelectric crystal and both are housed in a casing.

Hence it is nothing but a spring-mass system where the piezoelectrical crystal is acting as a spring.

It is always important to check the accelerometer specifications before selection of an accelerometer for your machine vibration measurement (refer Chapter 11). Few typical specifications are listed here.

- Sensitivity: pC/g, pC/(m/s2), mV/g, mV/(m/s2).
- Frequency range of measurement
- Resonance or natural frequency
- Maximum acceleration measurement limit
- Amplitude linearity—This is the ratio between the measured acceleration to the actual acceleration.
- Phase distortion—There should not be any phase error between the measured acceleration and the actual acceleration of the vibrating object.

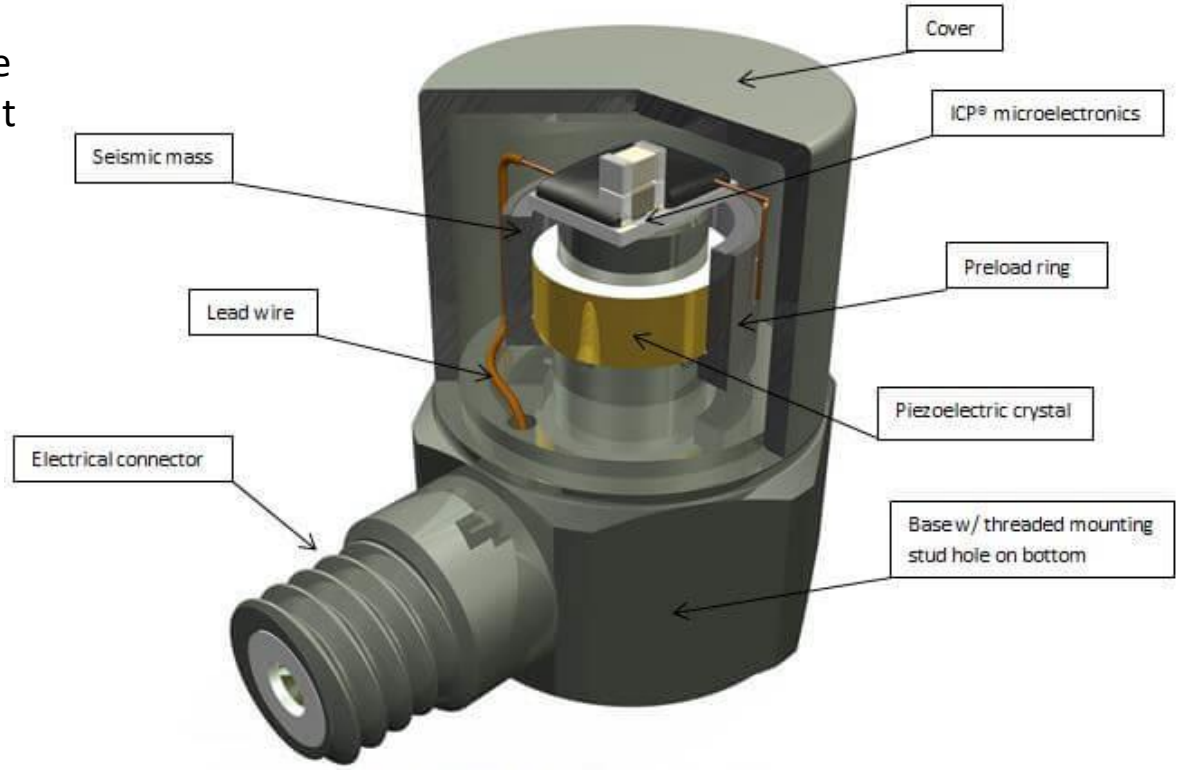


Figure 1: Typical ICP® Accelerometer

Mounting of Accelerometers

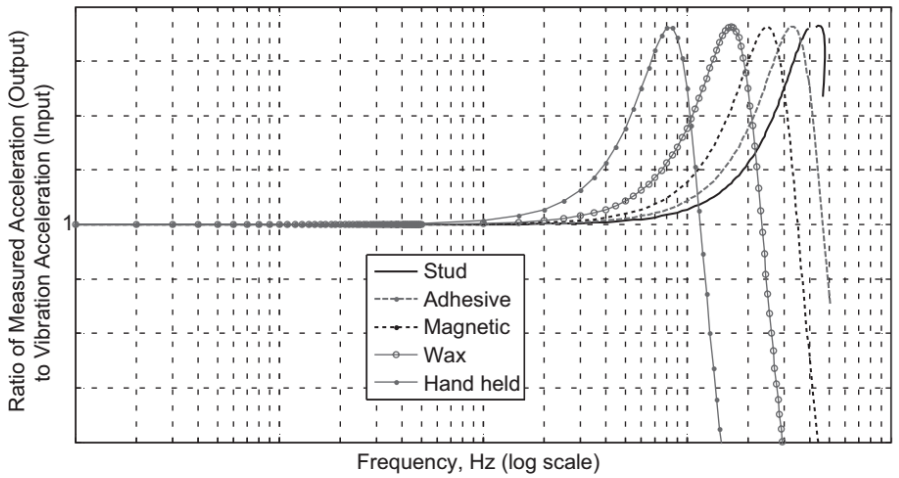


FIGURE 4.9 Impact of different mountings on accelerometer behavior.

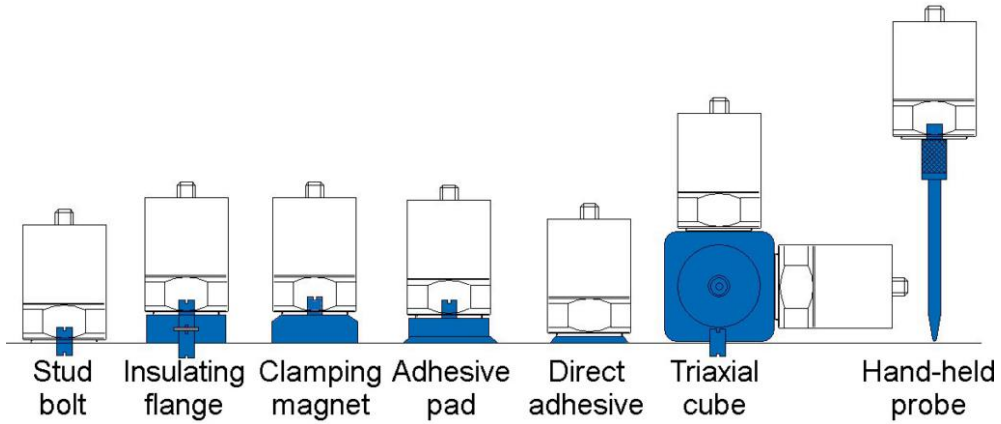


TABLE 4.1
Typical List of Accelerometer Specifications

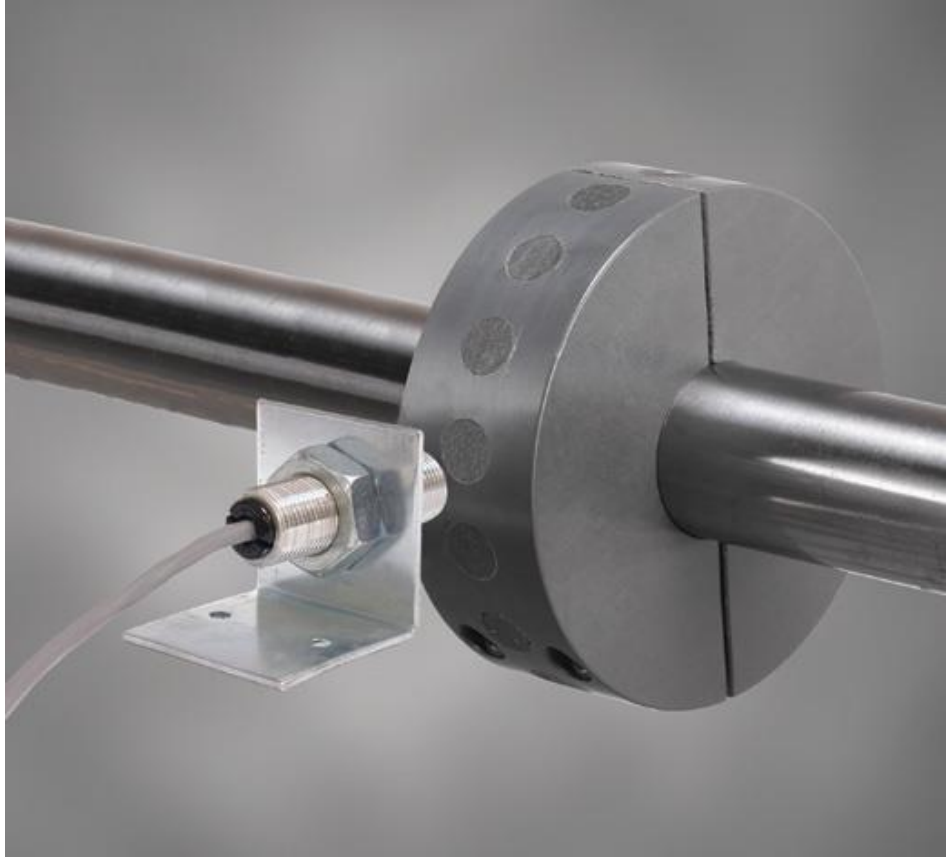
Parameters	Values	Remarks
Sensitivity	100 mV/g	Generates 100 mV for 1 g (or 9.81 m/s) vibration acceleration input.
Measurement range	±50 g	Limits of maximum and minimum level of vibration acceleration measurement.
Resonance frequency	≥50 kHz	This is the natural frequency of the accelerometer. For exact value, it is better to ask for test/calibration chart.
Frequency range (±5%)	0.5 Hz–10 kHz	10 kHz is acceptable as per the theory of 0.20 f_n but it can have amplitude error up to ± 5%.
Resolution	0.00015 g	This is the least count for the accelerometer i.e., the minimum fluctuation in vibration can be measured accurately.
Overload limit (shock)	±5000 g (peak)	Accelerometer may not get damaged up to this load.
Operating temperature range	−65° to +200°F	Accelerometers with different temperature range are also available commercially to meet industrial requirements.
Excitation voltage	18–30 V (DC)	Either a separate power unit for this accelerometer is need or the DAQ device should have this option.
Constant excitation current	2–20 mA	
Size	Select as per the requirements.	
Weight		

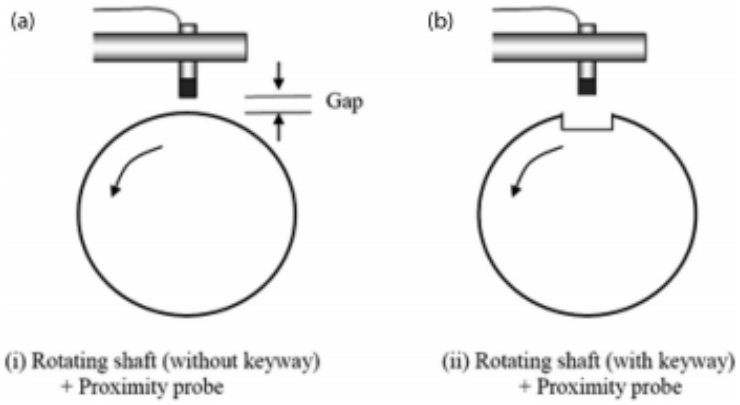
Tacho Sensor

In a rotating machine, a reference signal is required to:

- measure rotating speed
- the phase of rotating shaft and its vibration with respect to the reference signal.

The reference signal measured by a sensor is call tacho sensor. The proximity probe and laser pointer sensor are the most commonly used tacho sensors.





(i) Rotating shaft (without keyway) + Proximity probe

(ii) Rotating shaft (with keyway) + Proximity probe

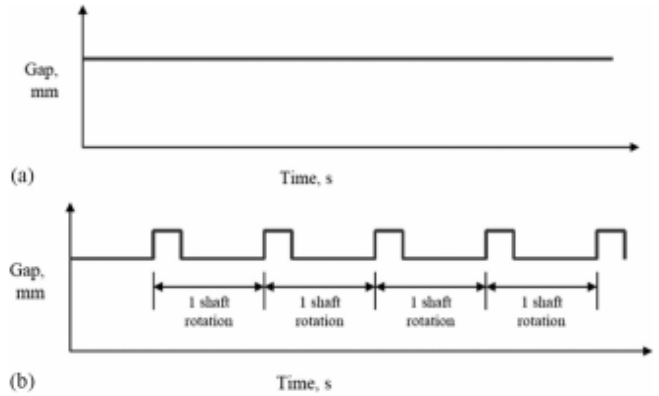


FIGURE 4.11 Measurement by the tacho sensor (a) without keyway (b) with keyway on the shaft.

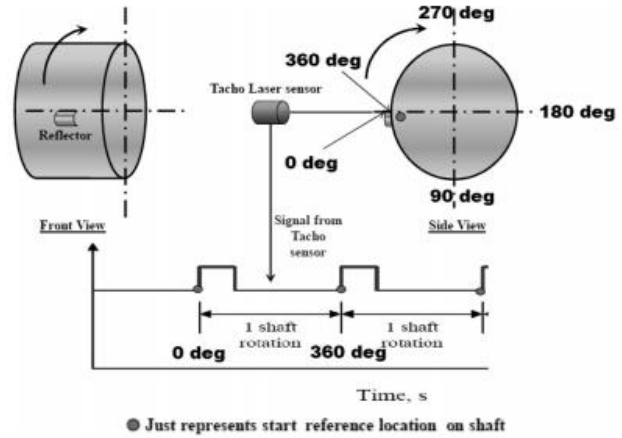


FIGURE 4.13 Speed and phase angle of the shaft with respect to the tacho sensor.



Section Two

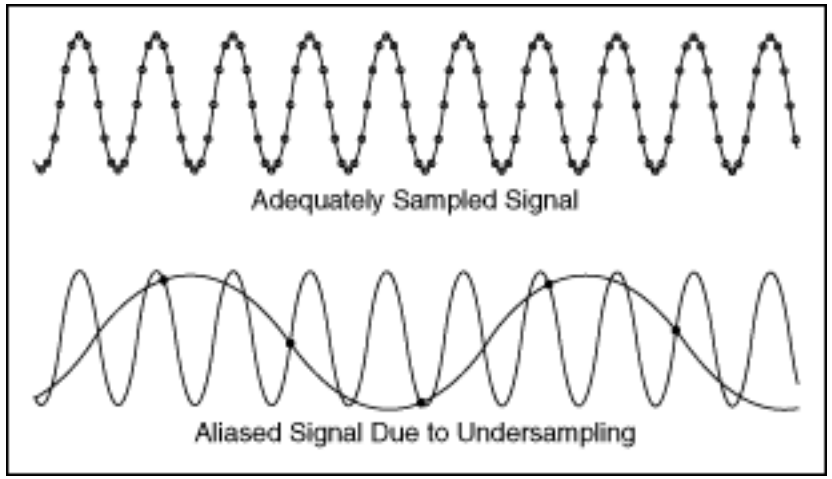
We will cover Signal Acquisition Fundamentals

- Fundamental
- ADC Sensitivity
- Bits in ADC

Signal Acquisition Fundamentals

Sampling Frequency

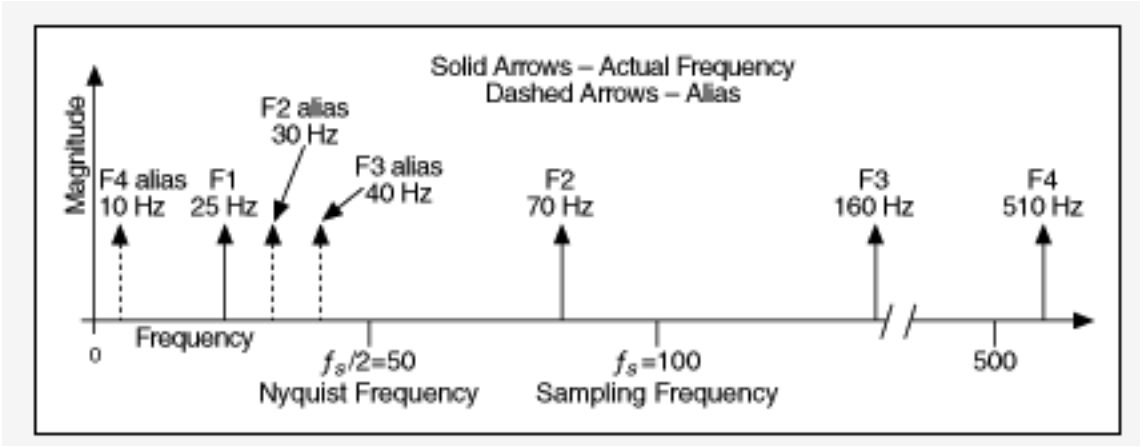
data sampling rate or sampling frequency, fs. : A definite rate of data collection needs to be defined, e.g., 20 samples/s. This means that 20 data samples from the signal will be collected in a second.



Aliasing Effect

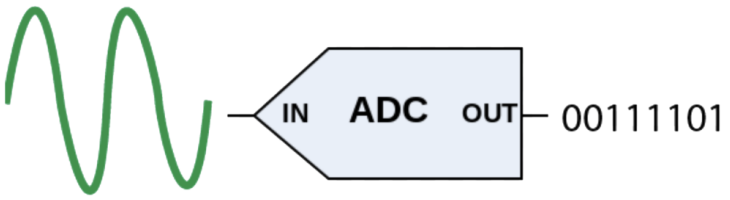
An aliased signal provides a poor representation of the analog signal. Aliasing causes a false lower frequency component to appear in the sampled data of a signal. $f_s - f_0$.

For a given sampling frequency, the maximum frequency you can accurately represent without aliasing is the Nyquist frequency.





Signal Acquisition Fundamentals



- **Nyquist Frequency**
  - Signal will be collected correctly only up to $f_s/2$ if the sampling frequency is fs samples per second (Hz) during the ADC process.
  - DAQ device must be equipped with a low pass filter, which can filter out all frequencies above the Nyquist frequency, fq during the data collection.
  - This low pass filter is known as the anti-aliasing filter.

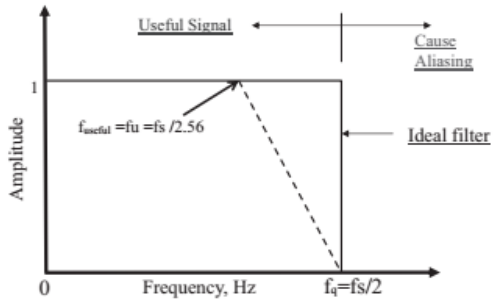


FIGURE 4.29 A simple representation of a standardized anti-aliasing filter used in practice.

- **Analog to Digital (ADC)**
- The ideal DAQ card should be capable of supporting the required number of input channels and the sampling frequency together with anti-aliasing filter per input channel for the simultaneous collection of data from all sensors.

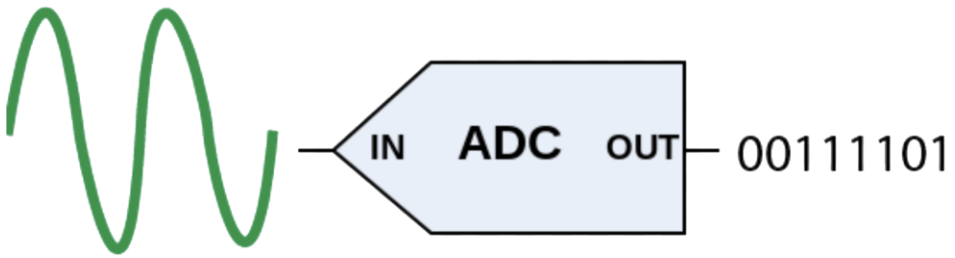
ADC Sensitivity

which is calculated as BT = 2^B. This means that BT = 2^4 = 16 binary bins for 4-bit DAQ device.

he sensitivity of the DAQ device is defined as the ratio of the full-scale input voltage (FSIV) to the total number of bins (BT).

$$S_d = \frac{FSIV}{BT}, \mathrm{BT} = 2^b$$

Consider a 16-bit DAQ card with two input channels. The input voltage level for each channel is assumed to ±5 V. Hence the sensitivity (S_d) of the DAQ card is $\frac{FSIV}{BT} = \frac{10}{2^{16}} = 0.15258$ mV/bin.



Bits in ADC and Resolutions

It is clear from these three cases the accuracy in the collected signals is much better in case of 24-bit DAQ device.

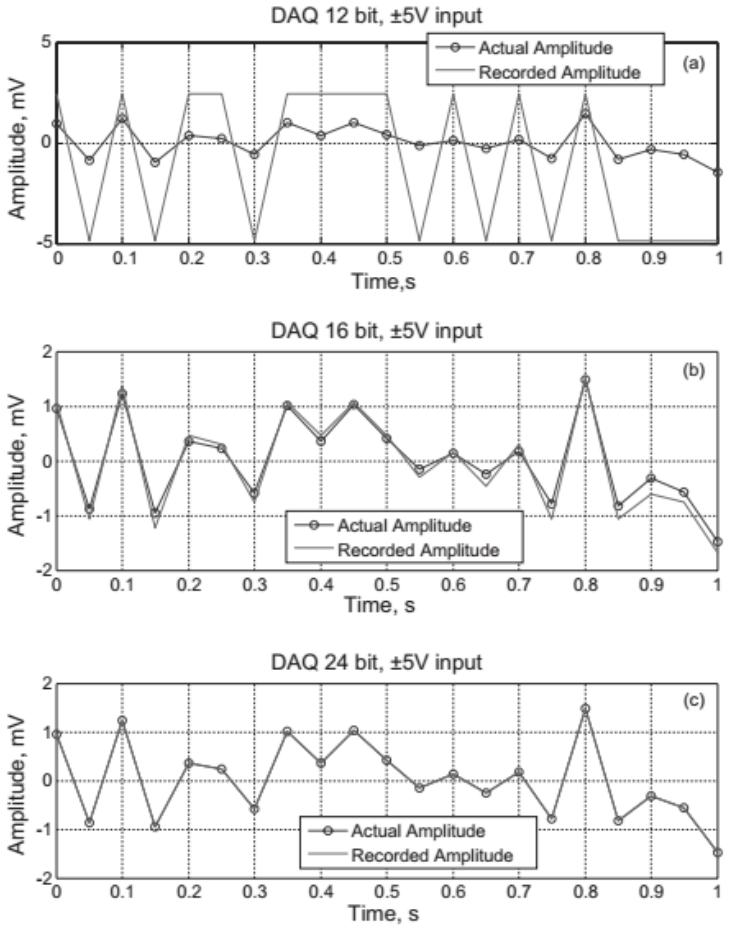
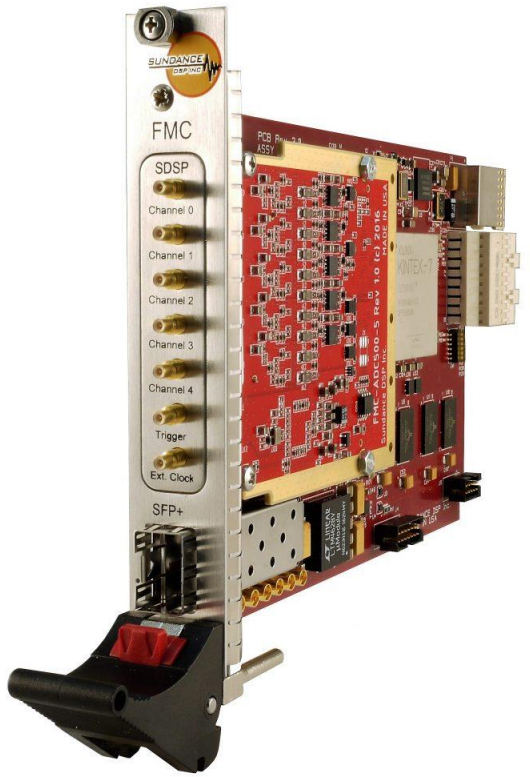


FIGURE 4.30 Recorded signal amplitudes with different DAQ devices (a) 12 bit, (b) 16 bit and (c) 24 bit.

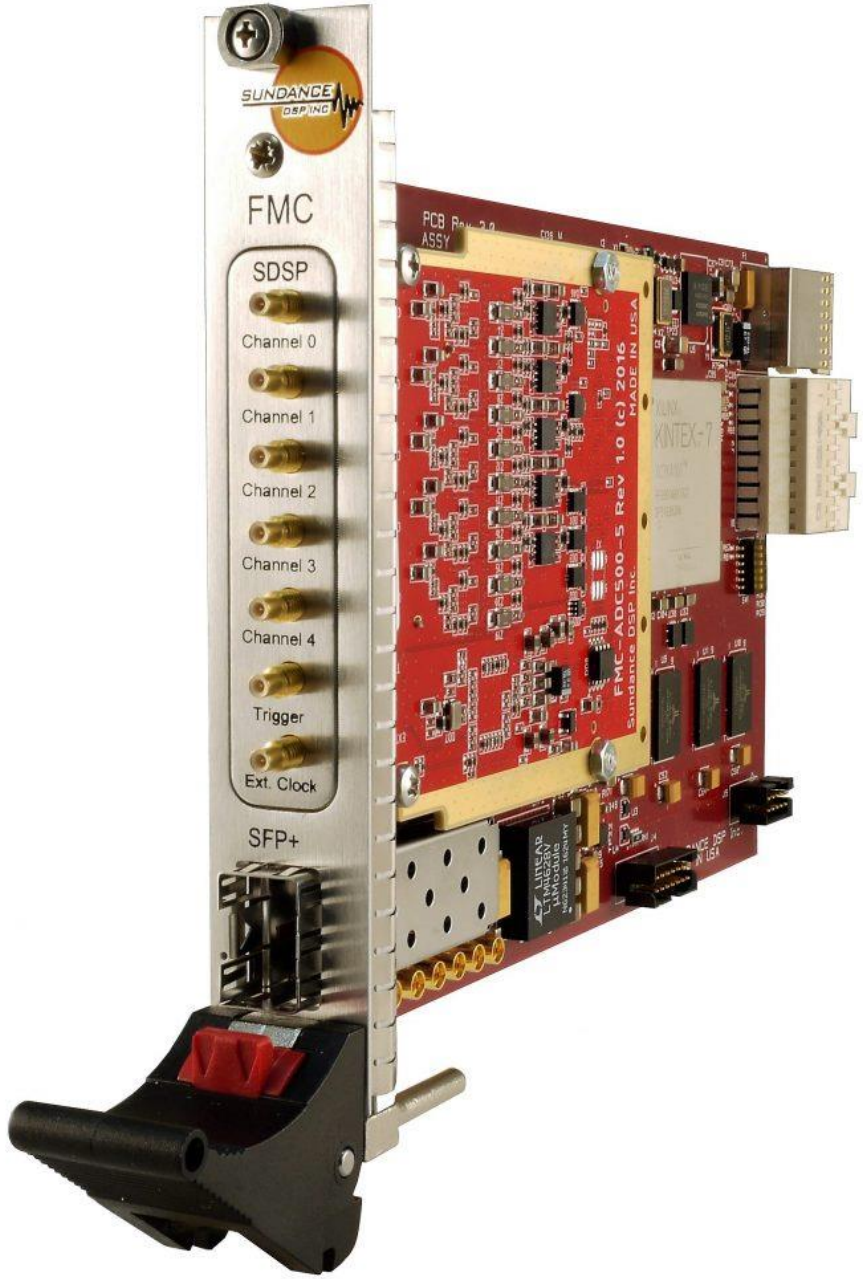


ADC Sensitivity Example

Vibration at 1m/s2 is measured using accelerometer with sensitivity of 100mv/g. Vibration signal recorded on 16bit DAQ with input voltage +-5V

Sensor sensitivity: S=100mV/g =0.1 V/g = (0.1/9.81) V/ms-2
Sensor output, S0 = P x S = 1 x (0.1/9.81) =0.0102 V
DAQ sensetivity

Sensor sensitivity, $S = 100$ mV/g $= 0.1$ V/g $= (0.1/9.81)$ V/m/s^2
Sensor output, $S_0 = P \times S = 1 \times (0.1/9.81) = 0.0102$ V;
DAQ sensitivity, $S_d = \frac{FSV}{2^b} = 0.15259$ mV/bin
Number of bins, $B = \frac{S_0}{S_d} = 66.84 = 67$ (integer number above 66)
Recorded voltage, $V_0 B \times S_d = 67 \times 0.15259$ mV $= 0.01022$ V





Thank you

Office hours:

Please send all questions through course messages